

Chapter 37

37 Producing Useful Work

2.4.2, 2.4.3

So far we have talked about doing work on gas in cylinders. That may not seem all that useful. But if we think that car engines have gas in cylinders with pistons, then it may seem like this could be important.

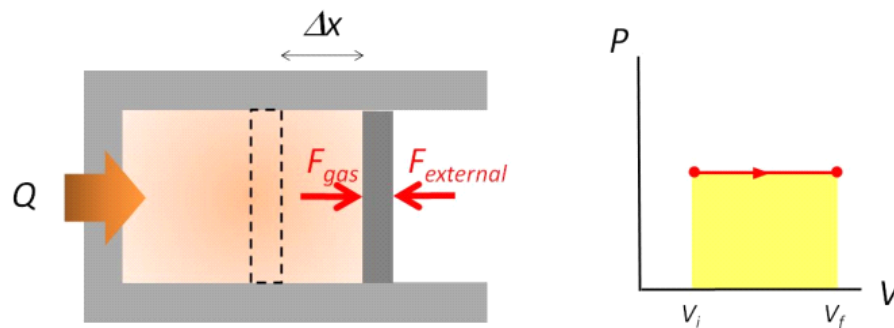
Fundamental Concepts

- $W_{\text{useful}} = -W_{\text{on the gas}}$
- A heat reservoir is a source of thermal energy that can be tapped without causing any measurable change in temperature.
- Energy Transfer Diagrams
- The Second Law of Thermodynamics
- Heat Engines
- Refrigerators
- Efficiency of a heat engine, $\eta = \frac{W_{\text{eng}}}{|Q_h|}$
- Coefficient of Performance (COP) for refrigerators
- To describe a heat engine cycle you need to find $P, V, n, T, \Delta E_{\text{int}}, Q, W$, for each part of the cycle and W_{eng}, Q_h, Q_c for the entire cycle.
- The Otto Cycle

37.1 Useful work.

Question 123.17.1

Suppose that we want to produce useful work from a thermodynamic device. For building such a device we have a nomenclature issue.



Consider our piston. If we add energy by heat, and the piston is free to move, then we do an amount of work on the gas

$$w = -P(V_f - V_i)$$

This is a negative amount of work, since V_f is larger than V_i . We defined work done on the gas as positive if the gas was being compressed, and the gas is not being compressed. So the work done on the gas is negative. But we could also define the work done by the gas. That is the negative of the work done on the gas.

$$w_{\text{on the gas}} = -w_{\text{by the gas}}$$

But think of a steam engine. We want the force by the gas on the piston, because this pushing of the gas on the piston is what we can use to push the wheel of a steam engine, or a turbine, etc. That is negative work on the gas. And it is inconvenient to have the useful work be negative. Let's define the useful work done by a thermodynamic device as the negative of work done on the gas.

$$w_{\text{useful}} = w_{\text{by the gas}} = -w_{\text{on the gas}}$$

But, w_{useful} is kind of a awkward name. Usually a machine that does useful work is called an engine, so let's call our useful work w_{eng} for "work done by an engine."

$$w_{\text{eng}} = -w$$

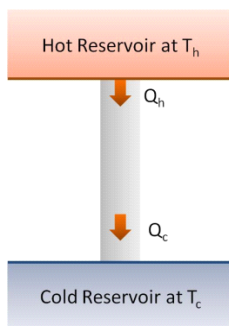
37.2 Reservoirs

Think of a water reservoir. In the western US we know what these are. It is a large man-made body of water that is designed to provide irrigation water for farms and homes.

Think of what happens when you water your lawn with water from the reservoir. Some water does leave from the reservoir, but compared to the whole reservoir of water, one lawn watering makes very little difference. Often there is more water coming into the reservoir, so really there is negligible water loss from your individual lawn.¹

A heat reservoir is like this. A hot reservoir is a source of thermal energy that can be tapped without causing any measurable change in temperature to the reservoir, itself. We can also have a cold reservoir that has a large lack of thermal energy. A small amount of thermal energy transferred to this cold reservoir won't measurably change its temperature.

The next figure is called an *Energy Transfer Diagram*.

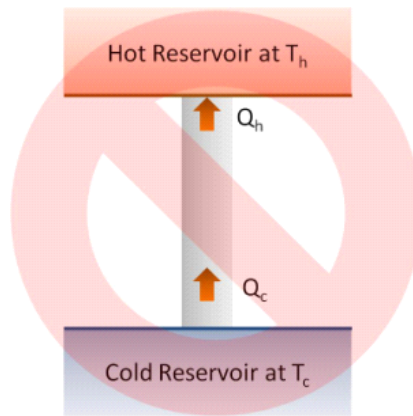


It shows energy moving by heat from a hot reservoir to a cold reservoir.

Using our understanding of the second law of thermodynamics, we can see that an amount of energy, Q_h , can flow from the hot reservoir and an amount of energy, Q_c , can flow by heat to the cold reservoir. We expect that if there is no other mechanism for dissipating the energy, then $Q_c = Q_h$. We will define both Q_h and Q_c as positive quantities, like vector magnitudes. If they are negative, we will explicitly write a negative sign.

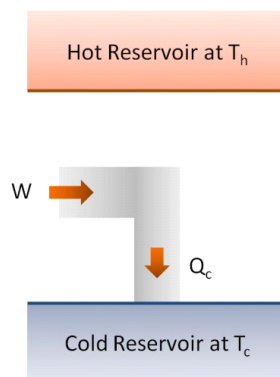
Look at this energy transfer diagram.

¹There is some danger in thinking of reservoirs this way. People at one time thought of the great lakes and oceans as reservoirs that were just too big for us to have any effect on them. But there are a lot of us, and if we all dump our trash in these reservoirs we can have a terrible effect on them. It has taken years to clean up the great lakes so they are safe to swim and safe to fish. The effect from one person is not enough to change the state of such a reservoir, but the effects from billions is enough to change its state. We should follow God's counsel and be good stewards of the Earth.



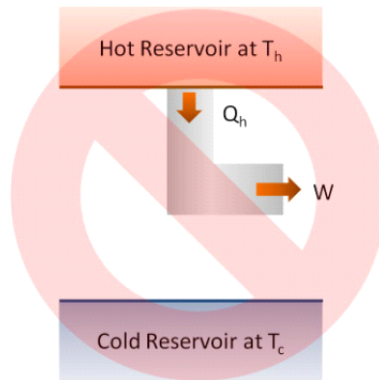
This diagram says that energy flows from the cold reservoir to the hot reservoir. From our experience, this doesn't happen. And our brand new Second Law of Thermodynamics says this won't happen! We only spontaneously experience energy coming from hot things and going to cold things. It would take work to make this energy transfer happen, but we don't have any work on the diagram. So this energy transfer diagram shows us a process that does not happen.

Here is another energy transfer diagram



This diagram says that we do have work, but that the work is directly converted into heat. This is a common experience. Rub your hands together. Your hands warm up. The work has created more internal energy, but that energy quickly dissipates into the room. The room temperature is not measurably affected by this increase in internal energy of the air. The air in the room is a cold reservoir.

Here is another diagram.



This diagram says that energy from the hot reservoir is converted directly into useful work. We expect that we could make a transfer of energy produce work. Think of our piston in the gas cylinder. But this diagram shows a 100% conversion of our energy transfer by heat becoming useful work. This is not something we expect to happen. Think about it, in a combustion engine we do get energy by heat by burning fuel, but we don't just get useful work (the car moves).

We also heat up the outside of the engine. If you touch your engine after it has run for a while you can burn your hand. Not all of the energy from the burning fuel became useful work. We know the reason for this already. We need to remove energy by heat to make our engine have a cyclic process so it doesn't melt down. Some of the energy from the burning fuel must leave to the cold reservoir as waste heat. But our diagram has no waste heat. This isn't possible.

We could define a generic device that takes energy by heat and converts it into useful work. We call such a device a *heat engine*. The heat engine described by the last diagram would be a perfect engine, converting all the energy transferred from the hot reservoir into useful work. And from our experience, this doesn't seem possible.

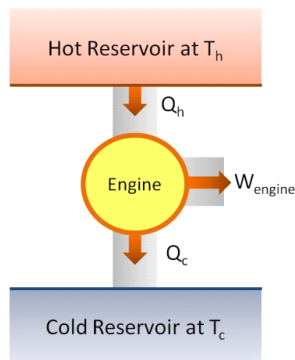
Our first law of thermodynamics told us about energy conservation, but it did not tell us what thermodynamic processes are possible. We need another law of thermodynamics that tells us what process we can expect to work. We have found two that don't work. So which processes can work?

Our Clausius' statement of the second law of thermodynamics told us one of these engine designs didn't work. And this is the job of the second law of thermodynamics, to tell us what to expect will actually happen. In terms of heat engines, we could add a new statement of our second law:

Second Law of Thermodynamics: Perfect engines are not possible

37.3 Heat Engines

Using our energy transfer diagram, let's introduce a more practical engine. Here is the diagram. We have added the engine explicitly on the diagram.



The engine takes energy from the heat reservoir and turns it into useful work. But there is some energy that is wasted and this energy ends up in the cold reservoir. Using our first law of thermodynamics, the amount of energy that becomes work must be

$$W_{\text{engine}} = Q_h - Q_c$$

We call $Q_h - Q_c$ the net heat

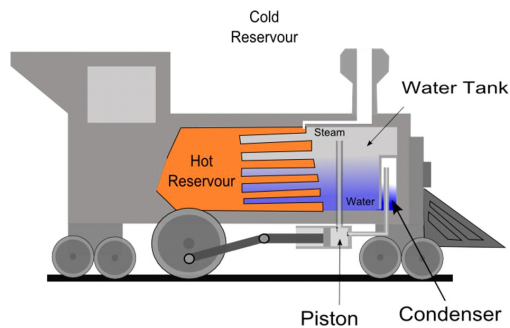
$$Q_{\text{net}} = Q_h - Q_c$$

Notice that Q_h is positive *for the engine* and Q_c is negative *for the engine*.

A useful heat engine is a device that takes in energy by heat and operates in a cycle. If it did not periodically return to its original state, then it could not keep going. The basics of the cycle are as follows

1. The working substance absorbs energy by heat from the high temperature reservoir
2. Work is done by the engine
3. Energy is expelled by heat to a low temperature reservoir

37.3.1 Example: Steam engine



In a steam engine

1. Water in the boiler absorbs energy from burning fuel (hot reservoir) and evaporates to steam
2. The steam does work by expanding against a piston
3. The steam cools and condenses releasing heat to the outside air (cold. reservoir) , and the liquid water returns to the boiler.

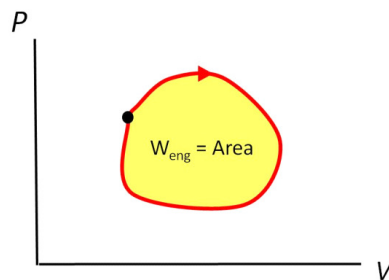
Since the engine goes through a cyclical process,

$$\Delta E_{int} = 0$$

for the water in the tank. Its initial and final internal energies are the same, so

$$Q_{net} = W_{eng} \quad (37.1)$$

The work done by the engine equals the *net* energy absorbed by the engine. The work is equal to the area enclosed by the curve of the PV diagram.



Thermal efficiency is defined as the ratio of the net work done by the engine during one cycle to the energy input at the higher temperature

$$\eta = \frac{W_{eng}}{|Q_h|} \quad (37.2)$$

$$= \frac{Q_{net}}{Q_h} \quad (37.3)$$

$$= \frac{|Q_h| - |Q_c|}{Q_h} \quad (37.4)$$

$$= 1 - \frac{|Q_c|}{|Q_h|} \quad (37.5)$$

We can think of this as

$$\eta = \frac{\text{what you gain in work}}{\text{what you gave in energy by heat}} \quad (37.6)$$

Sometimes efficiency is given the symbol e , but this is easy to mistake for the base of the natural logarithm, so let's use η .

For a car internal combustion engine

$$\eta \approx 20\% \quad (37.7)$$

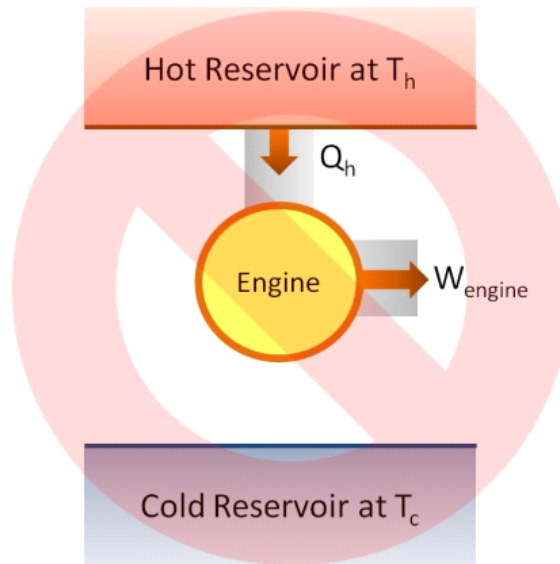
is a very good number (most student and faculty cars are far less efficient than this).

Suppose that we have a 100% efficient engine. We know that this is not possible, but consider what this would mean.

$$\eta = 1 - \frac{|Q_c|}{|Q_h|} \quad (37.8)$$

$$= 1 \quad (37.9)$$

this can only happen when $|Q_c| = 0$ which means that no energy is transferred out of the engine by heat – which is not really possible. The fact that real engines have efficiencies much lower than 1 leads to another statement of the second law of thermodynamics.



Second Law of Thermodynamics: It is impossible to construct a heat engine that, operating in a cycle, inputs energy by heat from a hot reservoir and converts the energy entirely into useful work.

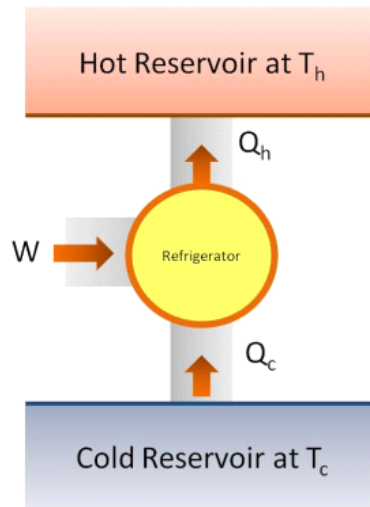
This is called the Kelvin-Planck form of the second law, and it means that

$$W_{eng} < |Q_h| \quad (37.10)$$

so the device represented by our drawing above is impossible to achieve.

37.4 Heat Pumps and Refrigerators

Let's look at a special form of a heat engine, a head pump.



A heat pump is an engine run in reverse. We wish to pump heat from a cold reservoir to the heat reservoir. To do this we must do work *on* the heat pump. This is not a natural process, so by adding in work we are adding energy to make it happen.

A window air conditioner unit is a good example of a heat pump. It removes energy by heat from the inside of your apartment and transfers that energy to the hot outside of your apartment. To make this happen, you need mechanical device that does work

A refrigerator is another example of a heat pump. It transfers energy by heat from the cold interior to the warmer environment of your apartment kitchen. It would be great if we could transfer heat from our cold reservoir (room of a house or freezer compartment of a fridge) to a hot reservoir without doing work, but this is not possible. This fact was discovered by Clausius and lead to the Clausius' statement of the second law that we already know, but we can update it so it is in terms of engines:

Second Law of Thermodynamics: It is impossible to construct a cyclical machine whose sole effect is to transfer energy continuously by heat from one object to another object at a higher temperature without the input of energy by work.

Or more simply, as we have already said: Energy does not transfer spontaneously by heat from a cold object to a hot object.

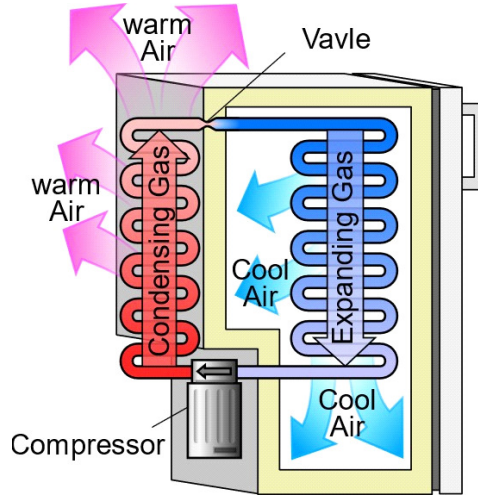
37.4.1 Coefficient of Performance

How well a heat pump works can be described by the *coefficient of performance* (COP).

$$COP_h = \frac{\text{energy transferred at high temperature}}{\text{work done on heat pump}} \quad (37.11)$$

$$= \frac{|Q_h|}{W} \quad (37.12)$$

COP is similar to efficiency. Think of our air conditioner. Q_h is typically higher than W . That is, we usually transfer more energy by heat to the outside world than we provide in work. The outside part of the air conditioner does indeed get hotter than the outside air temperature. That is why energy will transfer by heat. Values of *COP* are generally greater than 1, though it is possible for them to be less than 1. Of course we would like the *COP* to be as high as possible



Refrigerator Design. Red = Gas at high pressure and very high temperature. Pink = Liquid at high pressure and high temperature. Blue = Liquid at low pressure and very low temperature. Light Blue = Gas at low pressure and low temperature.

(Drawing courtesy Jahobr)

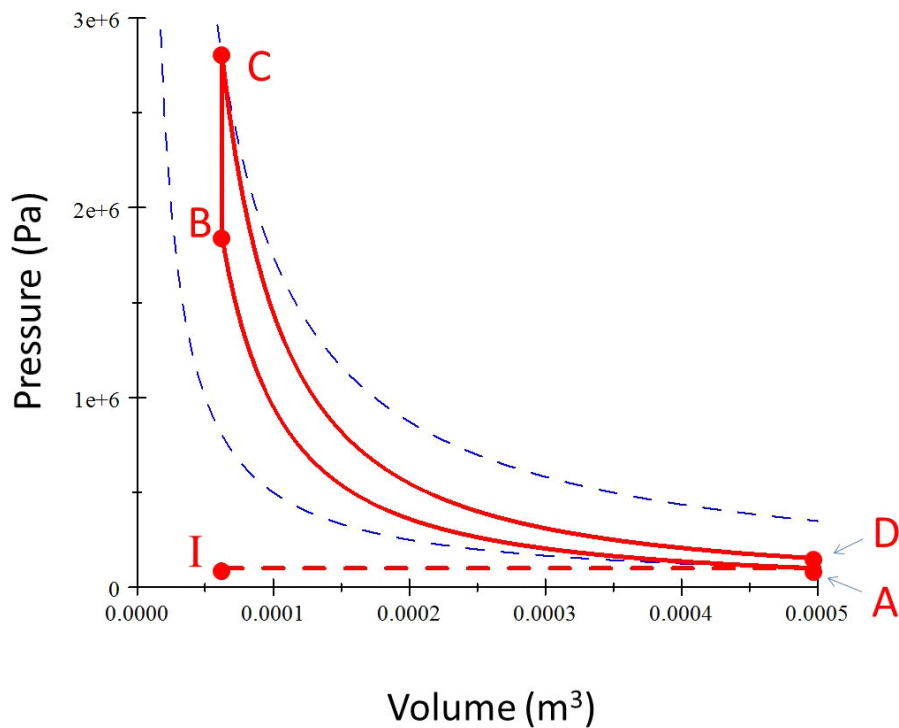
For refrigeration, the important thing is how much energy we transfer out of the cool area, Q_c , so for cooling the *COP* is

$$COP_c = \frac{|Q_c|}{W} \quad (37.13)$$

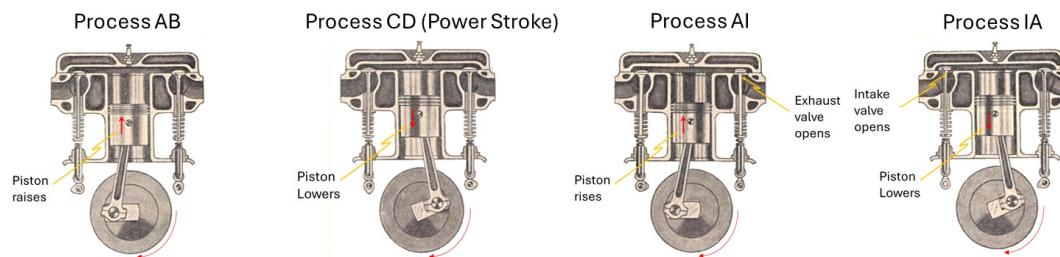
A good refrigerator should have a high COP, Typical values are 5 or 6.

37.5 An example: The Otto Cycle

But so far we have been very vague about what happens in the yellow engine circle in our energy transfer diagrams. Let's look at a (mostly) real example of a heat engine. This model of a heat engine is due to a researcher named Otto, so it is called the *Otto Cycle*.



The PV diagram above shows the Otto cycle. This cycle is approximately what happens in a gasoline engine, like those in cars or, more like what we find in lawn mowers. It consists of four main processes. Those processes correspond to a piston moving in a cylinder (our favorite thermodynamic system!). Here is a picture of what is happening in the engine.



To understand the Otto cycle let's do our job of finding ΔE_{int} , Q , and w for each part of the cycle and then describe the state of the gas at the beginning and end of each cycle by finding P , V , n , T , and E_{int} .

For our engine, suppose the compression ratio is

$$\frac{V_A}{V_B} = 8.00$$

Let's take

$$\begin{aligned} V_A &= 500 \text{ cm}^3 = 0.0005 \text{ m}^3 \\ P_A &= 100 \text{ kPa} \\ T_A &= 293 \text{ K} \end{aligned}$$

Then we can find V_B

$$V_B = \frac{V_A}{8} = 62.5 \text{ cm}^3 = 6.25 \times 10^{-5} \text{ m}^3$$

and we will need to measure the temperature at C . Suppose we measure

$$T_C = 1023 \text{ K}$$

and the intake temperature to be

$$T_I = 273 \text{ K}$$

(which probably tells us that this must be a road trip in Rexburg, not Phoenix). Further suppose our engine uses a diatomic ideal gas with

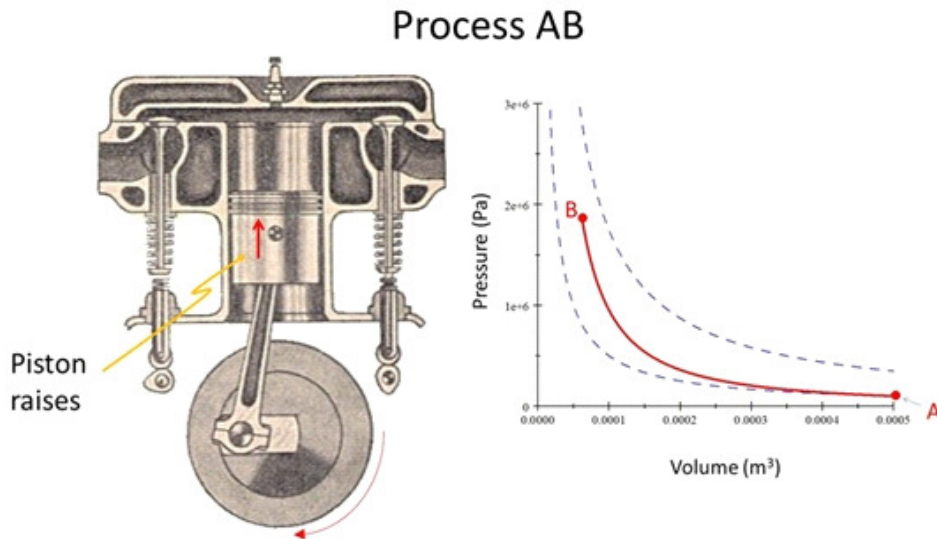
$$\begin{aligned} C_V &= \frac{5}{2}R \\ R &= 8.314 \frac{\text{J}}{\text{mol K}} \end{aligned}$$

so

$$\gamma = 1.4$$

Let's follow the process through to see what the state variables are at each point on the PV diagram, and what Q_h and Q_c are for each process, and what the

work, W , is and what the change in internal energy will be, ΔE_{int} . Finally, we will find the efficiency of the engine.



Process $A \rightarrow B$ is an adiabatic compression. The piston moves upward, compressing the fuel-air mixture. This happens rapidly, so there is no time for energy to leave by heat, $Q = 0$. In this process $\Delta E_{int} = W$.

We can use

$$P_i V_i^\gamma = P_f V_f^\gamma$$

to find the final pressure for this process at B

$$P_B = P_A \frac{V_A^\gamma}{V_B^\gamma}$$

But of course this works for any point on the line from A to B so

$$P = P_A \frac{V_A^\gamma}{V^\gamma}$$

We can use this form for P to find the work for process $A \rightarrow B$

$$\begin{aligned}
 w_{AB} &= - \int_{V_A}^{V_B} P dV \\
 &= - \int_{V_A}^{V_B} P_A \frac{V_A^\gamma}{V^\gamma} dV \\
 &= -P_A V_A^\gamma \int_{V_A}^{V_B} \frac{1}{V^\gamma} dV \\
 &= -P_A V_A^\gamma \left(\frac{V^{1-\gamma}}{1-\gamma} \right) \Big|_{V_A}^{V_B} \\
 &= -\frac{P_A V_A^\gamma}{1-\gamma} (V_B^{1-\gamma} - V_A^{1-\gamma})
 \end{aligned}$$

We can rearrange this into a more convenient formula

$$\begin{aligned}
 w_{AB} &= \frac{P_A V_A^\gamma}{\gamma-1} \left(\frac{1}{V_B^{\gamma-1}} - \frac{1}{V_A^{\gamma-1}} \right) \\
 &= \frac{1}{\gamma-1} \left(\frac{P_A V_A^\gamma}{V_B^{\gamma-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right) \\
 &= \frac{1}{\gamma-1} \left(\frac{P_A V_A^\gamma}{V_B^\gamma V_B^{-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right)
 \end{aligned}$$

And once again using

$$P_B V_B^\gamma = P_A V_A^\gamma$$

or

$$P_A = P_B \frac{V_B^\gamma}{V_A^\gamma}$$

We get

$$w_{AB} = \frac{1}{\gamma-1} \left(\frac{P_B \frac{V_B^\gamma}{V_A^\gamma} V_A^\gamma}{V_B^\gamma V_B^{-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right)$$

A lot of things cancel. We get

$$\begin{aligned}
 w_{AB} &= \frac{1}{\gamma-1} \left(\frac{P_B}{V_B^{-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right) \\
 &= \frac{1}{\gamma-1} (P_B V_B - P_A V_A)
 \end{aligned}$$

which we can generalize into another basic equation for adiabatic processes!

$$w_{if} = \left(\frac{1}{\gamma-1} \right) (P_f V_f - P_i V_i)$$

Then for process $A \rightarrow B$ we have

$$w_{AB} = \left(\frac{1}{\gamma - 1} \right) (P_B V_B - P_A V_A)$$

and we can use our equation for P_B that we got before, $P_B = P_A \frac{V_A^\gamma}{V_B^\gamma}$ so that our w_{AB} equation becomes

$$w_{AB} = \left(\frac{1}{\gamma - 1} \right) \left(P_A \frac{V_A^\gamma}{V_B^\gamma} V_B - P_A V_A \right)$$

And from our compression ration equation

$$V_B = \frac{V_A}{8.00}$$

so w_{AB} becomes

$$\begin{aligned} w_{AB} &= \left(\frac{1}{\gamma - 1} \right) \left(P_A \frac{V_A^\gamma}{\left(\frac{V_A}{8} \right)^\gamma} \frac{V_A}{8.00} - P_A V_A \right) \\ &= \left(\frac{1}{\gamma - 1} \right) \left(8^\gamma P_A \left(\frac{V_A}{8} \right) - P_A V_A \right) \\ &= \left(\frac{V_A}{\gamma - 1} \right) (8^{\gamma-1} P_A - P_A) \\ &= \left(\frac{V_A P_A}{\gamma - 1} \right) (8^{\gamma-1} - 1) \end{aligned}$$

So the work is

$$w_{AB} = \left(\frac{V_A P_A}{\gamma - 1} \right) (8^{\gamma-1} - 1)$$

and we know all of these numbers so

$$\begin{aligned} w_{AB} &= \left(\frac{(0.0005 \text{ m}^3) (100 \text{ kPa})}{1.4 - 1} \right) (8^{1.4-1} - 1) \\ &= 162.17 \text{ J} \end{aligned}$$

But this is the work done on the gas, the useful work is the negative of this

$$w_{engAB} = -162.17 \text{ J}$$

which tells us that it took energy to make the process $A \rightarrow B$ happen. You might need a pull cord to start this engine or an electric starter motor. The change in internal energy is

$$\Delta E_{AB} = 162.17 \text{ J}$$

and we can find

$$E_{int} = \frac{5}{2}nRT$$

but we need n . We can get it from what we know at A using the ideal gas law

$$\begin{aligned} n &= \frac{P_A V_A}{RT_A} \\ &= \frac{(100 \text{ kPa}) (0.0005 \text{ m}^3)}{\left(8.314 \frac{\text{J}}{\text{mol K}}\right) (293 \text{ K})} \\ &= 2.0525 \times 10^{-2} \text{ mol} \end{aligned}$$

Then

$$\begin{aligned} E_A &= \frac{5}{2}nRT_A \\ &= \frac{5}{2} (2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right) (293 \text{ K}) \\ &= 125.00 \text{ J} \end{aligned}$$

We used

$$P_B = P_A \frac{V_A^\gamma}{V_B^\gamma}$$

but we should find a numeric answer for P_B . Let's use our compression ratio equation for V_B again

$$\begin{aligned} P_B &= P_A \frac{V_A^\gamma}{\left(\frac{V_A}{8}\right)^\gamma} \\ &= P_A \frac{V_A^\gamma}{\frac{V_A^\gamma}{8^\gamma}} \\ &= 8^\gamma P_A \\ &= 8^{1.4} (100 \text{ kPa}) \\ &= 1837.9 \text{ kPa} \end{aligned}$$

And finally, from the ideal gas law

$$\begin{aligned} T_B &= \frac{P_B V_B}{nR} \\ &= \frac{(1837.9 \text{ kPa}) (6.25 \times 10^{-5} \text{ m}^3)}{(2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right)} \\ &= 673.15 \text{ K} \end{aligned}$$

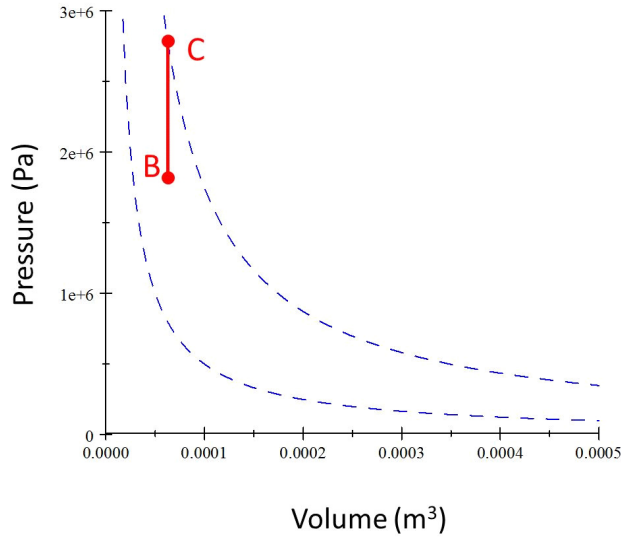
At last we can find E_B

$$\begin{aligned} E_B &= \frac{5}{2}nRT_B \\ &= \frac{5}{2} (2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right) (673.15 \text{ K}) \\ &= 287.17 \text{ J} \end{aligned}$$

what we have learned is summarized in this table

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
A	293	100	0.000 5	$2.052\,5 \times 10^{-2}$	125.00	—
B	673.15	1837.9	6.25×10^{-5}	$2.052\,5 \times 10^{-2}$	287.17	—
Process	Q (J)	W _{int} (J)	ΔE _{int} (J)	W _{eng} (J)	Q _h	Q _c
AB	0	162.17 J	162.17 J	−162.17 J	0	0

That was just one process!



Process $B \rightarrow C$ is an isochoric process. This is where the spark plug ignites the air-fuel mixture created by the fuel injectors. A great deal of energy is released in a hurry. The temperature jumps up and the pressure does too, but the volume is the same because this process happens in a flash—literally!

So this must be an isochoric process. We expect $W_{BC} = 0$, so $\Delta E_{BC} = Q_{BC}$. The pressure increases, so we expect Q_{BC} to be positive. This must be part of Q_h ! We know the temperature at C and we know the volume $V_C = V_B$ so we can find the final pressure

$$\begin{aligned}
 P_C &= \frac{nRT_C}{V_C} \\
 &= \frac{(2.052\,5 \times 10^{-2}) (8.314 \frac{\text{J}}{\text{mol K}}) (1023)}{6.25 \times 10^{-5} \text{ m}^3} \\
 &= 2.793\,1 \times 10^6 \text{ Pa} \\
 &= 2793.1 \text{ kPa}
 \end{aligned}$$

We can find the internal energy at C now

$$\begin{aligned} E_C &= \frac{5}{2}nRT_C \\ &= \frac{5}{2}(2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (1023 \text{ K}) \\ &= 436.42 \text{ J} \end{aligned}$$

We can use

$$\begin{aligned} Q &= nC_V\Delta T \\ &= n\frac{5}{2}R(T_C - T_B) \end{aligned}$$

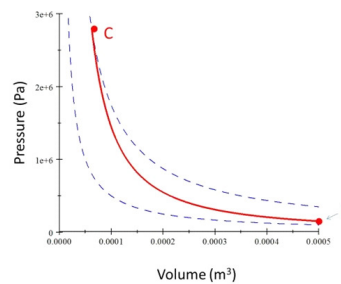
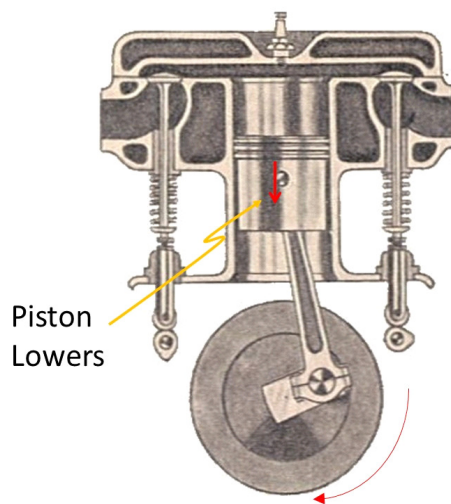
For the process BC to find the energy transfer by heat

$$\begin{aligned} Q_{BC} &= (2.0525 \times 10^{-2} \text{ mol}) \frac{5}{2} \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (1023 \text{ K} - 673.15 \text{ K}) \\ &= 149.25 \text{ J} \end{aligned}$$

We already mentioned that $Q_{BC} = \Delta E_{intBC}$ and $W_{BC} = 0$, so we can complete our table of state variables and process variables for this process.

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
B	673.15	1837.9	6.25×10^{-5}	2.0525×10^{-2}	287.17	—
C	1023	2793.1	6.25×10^{-5}	2.0525×10^{-2}	436.42	—
Process	Q (J)	W _{int} (J)	ΔE _{int} (J)	W _{eng} (J)	Q _h	Q _c
BC	149.25	0	149.25	0	149.25	0

Process CD (Power Stroke)



Process is $C \rightarrow D$ is another adiabatic process, this time an expansion. This is sometimes called the “power stroke” because the hot, high pressure gas pushes the piston down. This push is what makes the car or lawn mower go. We know that $Q_{CD} = 0$ so $\Delta E_{CD} = W_{CD}$

To find the ideal gas state variables at D let's start with

$$P_i V_i^\gamma = P_f V_f^\gamma$$

again, then

$$P_D = P_C \frac{V_C^\gamma}{V_D^\gamma}$$

but from our PV diagram we see $V_C = V_B$ and $V_D = V_A$. This makes sense because the cylinder and piston system maximum and minimum volumes don't change. So we can write

$$P_D = P_C \frac{V_B^\gamma}{V_A^\gamma} = 8^{-\gamma} P_C$$

and knowing this we can find the temperature at D using the ideal gas law

$$T_D = \frac{P_D V_D}{nR}$$

let's numerically calculate P_D and T_D at this point.

$$\begin{aligned} P_D &= 8^{-1.4} (2793.1) \\ &= 151.97 \text{ kPa} \end{aligned}$$

and

$$\begin{aligned} T_D &= \frac{(151.97 \text{ kPa}) (0.0005 \text{ m}^3)}{(2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right)} \\ &= 445.28 \text{ K} \end{aligned}$$

And while we are at it we can complete our state D by calculating E_D

$$\begin{aligned} E_D &= \frac{5}{2} n R T_D \\ &= \frac{5}{2} (2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right) (445.28 \text{ K}) \\ &= 189.96 \text{ J} \end{aligned}$$

Now for the work done, we again use our new adiabatic equation

$$\begin{aligned} W_{CD} &= \left(\frac{1}{\gamma - 1}\right) (P_D V_D - P_C V_C) \\ &= \left(\frac{1}{\gamma - 1}\right) (8^{-\gamma} P_C V_A - P_C V_B) \\ &= \left(\frac{P_C}{\gamma - 1}\right) (8^{-\gamma} V_A - V_B) \end{aligned}$$

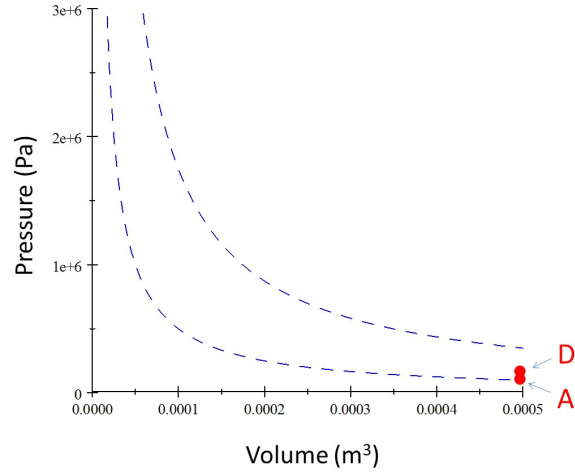
all of which we know, so

$$\begin{aligned} W_{CD} &= \left(\frac{2793.1 \text{ kPa}}{1.4 - 1} \right) (8^{-1.4} (0.0005 \text{ m}^3) - 6.25 \times 10^{-5} \text{ m}^3) \\ &= -246.46 \text{ J} \end{aligned}$$

which completes our set for process CD

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
C	1023	2793.1	6.25×10^{-5}	2.0525×10^{-2}	436.42	—
D	445.28	151.97	0.0005	2.0525×10^{-2}	189.96 J	—
Process	Q (J)	W _{int} (J)	ΔE _{int} (J)	W _{eng} (J)	Q _h	Q _c
CD	0	-246.46	-246.46	246.46	0	0

Process $D \rightarrow A$ is an isochoric process. We are nearing the end of the cycle, In process $D \rightarrow A$ a valve opens letting the pressure drop quickly.



Since this happens quickly, the piston does not have time to change position, so the volume does not change. Process DA is (almost) isochoric. We already know P_A , V_A , n , and T_A . But we need to find the process work, heat and change in internal energy. Since the volume does not change, $W_{DA} = 0$ and $\Delta E_{DA} = Q_{DA}$ which we can find using

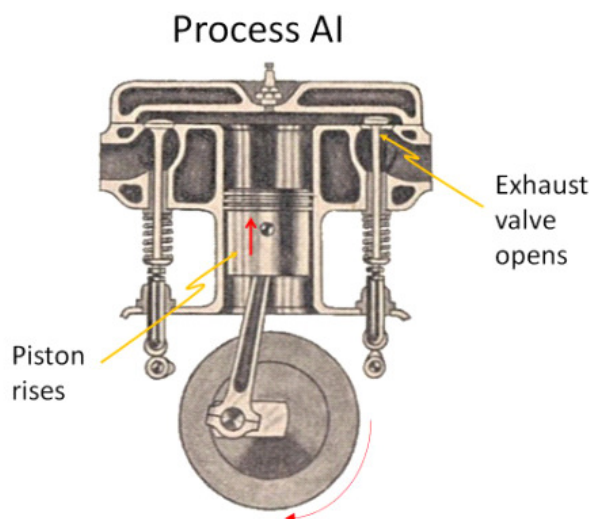
$$\begin{aligned} Q_{DA} &= nC_V\Delta T \\ &= n\frac{5}{2}R(T_A - T_D) \end{aligned}$$

which gives

$$\begin{aligned} Q_{DA} &= (2.0525 \times 10^{-2} \text{ mol}) \frac{5}{2} \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (293 \text{ K} - 445.28 \text{ K}) \\ &= -64.964 \text{ J} \end{aligned}$$

which tells us that we have lost energy by heat. So Q_{DA} must contribute to Q_c .

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
D	445.28	151.97	0.000 5	$2.052\,5 \times 10^{-2}$	189.96 J	—
A	293	100	0.000 5	$2.052\,5 \times 10^{-2}$	125.00	—
Process	Q (J)	W _{int} (J)	ΔE _{int} (J)	W _{eng} (J)	Q _h	Q _c
DA	−64.964	0	−64.964	0	0	64.964



But we are not done. The opening of the valve let the spent fuel begin to leave as exhaust. But we can help remove the exhaust by pushing the piston up. This is done at constant pressure, because the valve is open to the exhaust system air pressure. We will lose gas by venting it. We can find the new number of moles using the ideal gas law

$$P_I V_I = n_I R T$$

since $P_I = P_A$ and

$$\frac{V_A}{V_I} = 8.00$$

$V_I = 8/V_A$ so we can guess that we should have

$$\frac{P_A V_A}{P_I V_I} = \frac{n_A R T_A}{n_I R T_I}$$

$$8 = \frac{n_A T_A}{n_I T_I}$$

so

$$n_I = \frac{n_A T_A}{8 T_I}$$

or

$$\begin{aligned} n_I &= \frac{(2.0525 \times 10^{-2} \text{ mol})(293 \text{ K})}{8(273 \text{ K})} \\ &= 2.7536 \times 10^{-3} \text{ mol} \end{aligned}$$

and then E_I will be

$$\begin{aligned} E_D &= \frac{5}{2} n_I R T_I \\ &= \frac{5}{2} (2.7536 \times 10^{-3} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (273 \text{ K}) \\ &= 15.625 \text{ J} \end{aligned}$$

And the work done for an isobaric process is

$$w_{AI} = P_A (V_I - V_A)$$

but this assumed that n did not change. Really we are just moving the gas, so our work is

$$\begin{aligned} w &= F \Delta x \\ &\approx 0 \end{aligned}$$

because it takes very little force to push out the gas, so we can ignore w_{AI} .

Finding Q_{AI} we would like to use

$$Q = n C_P \Delta T$$

but we quickly run into trouble again because n changed! So what has gone wrong?

Well we are using convection to remove heat instead of conduction. We can get around this by finding ΔE_{int} directly.

$$\begin{aligned} \Delta E_{AI} &= E_I - E_A \\ &= 15.625 \text{ J} - 125.00 \text{ J} \\ &= -109.38 \text{ J} \end{aligned}$$

From this and the first law

$$\Delta E_{int} = Q + W$$

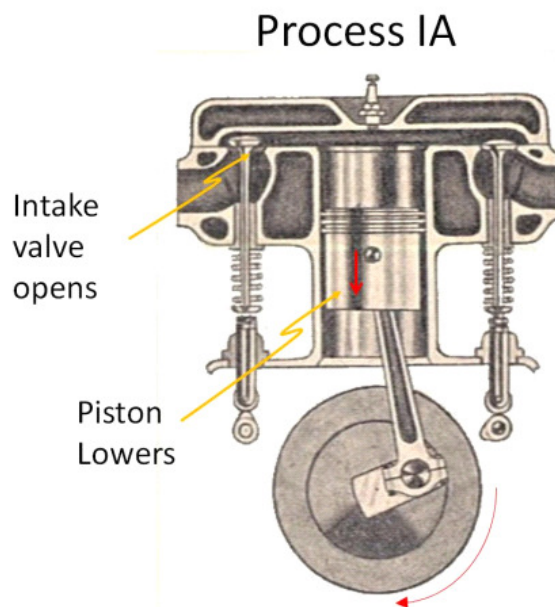
We can find Q

$$\begin{aligned} Q &\approx \Delta E_{int} \\ &\approx -109.38 \text{ J} \end{aligned}$$

Energy is leaving with the gas, so we seem to pick up a contribution to Q_c . But really this is not part of our thermodynamic process, because we mechanically removed the energy by removing the gas. Let's denote this by putting our

energy in parenthesis to remind us that this is a more mechanical than thermal process.

State	T (K)	P (kPa)	V (m^3)	n (mol)	E_{int} (J)	—
A	293	100	0.000 5	$2.052\,5 \times 10^{-2}$	125.00	—
I	273	100	6.25×10^{-5}	$2.753\,6 \times 10^{-3}$	15.625	—
Process	Q (J)	W_{int} (J)	ΔE_{int} (J)	W_{eng} (J)	Q_h	Q_c
AI	(-109.38)	~ 0	(-109.38)	~ 0	0	(109.38)



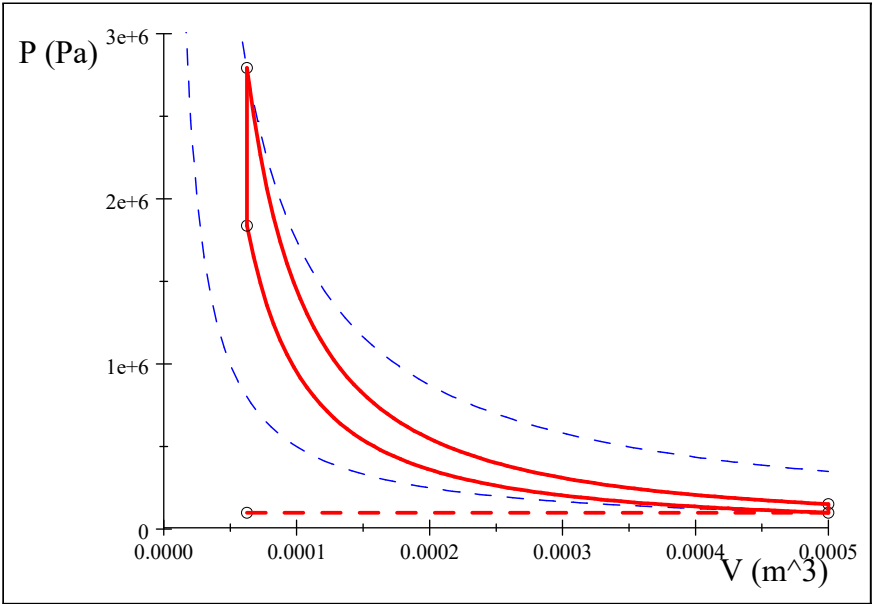
Now the exhaust valve closes, and the intake valve opens. The new fuel mixture is brought in. We essentially reverse process *AI* in process *IA*.

State	T (K)	P (kPa)	V (m^3)	n (mol)	E_{int} (J)	—
I	293	100	6.25×10^{-5}	$2.565\,7 \times 10^{-3}$	15.625	—
A	293	100	0.000 5	$2.052\,5 \times 10^{-2}$	125.00	—
Process	Q (J)	W_{int} (J)	ΔE_{int} (J)	W_{eng} (J)	Q_h	Q_c
IA	(109.38)	~ 0	(109.38)	~ 0	(109.38)	0

But there is a difference. The new gas has useful fuel. so we are ready to complete the cycle again by compressing the gas, igniting it, and receiving the power stroke and exhaust. Note that Processes *AI* and *IA* have a net zero effect thermodynamically, but are very important to making your car or lawn mower go!

Putting it all together gives our *PV* diagram for the whole cycle. Note that since n changed for the *AI* – *IA* processes, it is not really possible to include it

on the PV diagram. That is why it has a dashed line. We know the volume and pressure along the path, but the predicted temperature from the graph may not be right because n changed too.



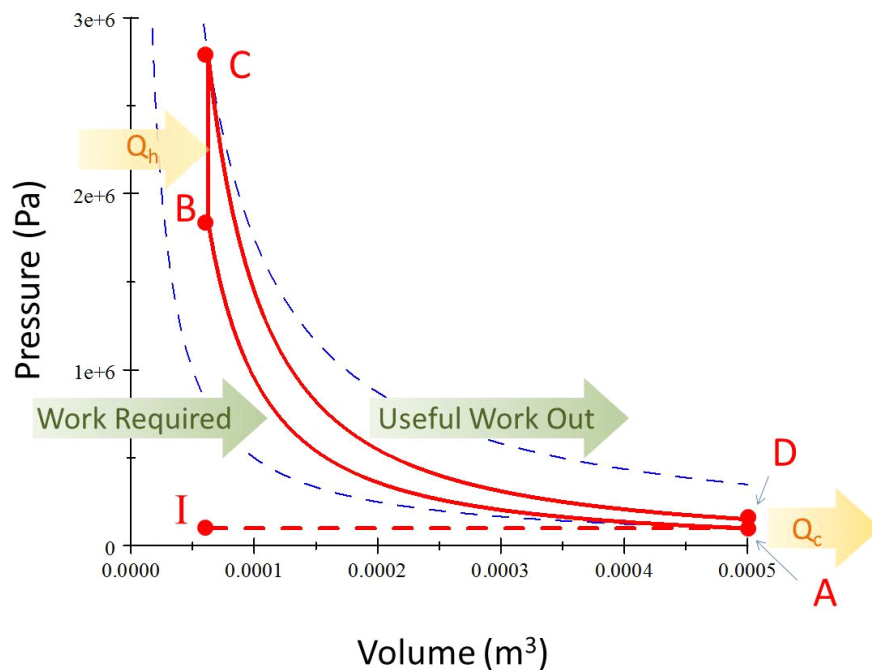
Let's summarize what we found in large tables. First the ideal gas state variables and E_{int}

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)
A	293	100	0.000 5	$2.052\,5 \times 10^{-2}$	125.00
B	673.15	1837.9	6.25×10^{-5}	$2.052\,5 \times 10^{-2}$	287.17
C	1023	2793.1	6.25×10^{-5}	$2.052\,5 \times 10^{-2}$	436.42
D	445.28	151.97	0.000 5	$2.052\,5 \times 10^{-2}$	189.96 J
I	293	100	6.25×10^{-5}	$2.565\,7 \times 10^{-3}$	15.625

and now the values of Q , W , and E_{int} for the processes. We have included W_{eng} , Q_h and Q_c as well

Process	Q (J)	W _{int} (J)	ΔE _{int} (J)	W _{eng} (J)	Q _h	Q _c
AB	0	162.17	162.17	-162.17	0	0
BC	149.25	0	149.25	0	149.25	0
CD	0	-246.46	-246.46	246.46	0	0
DA	-64.964	0	-64.964	0	0	64.964
AI	(-109.38)	(~ 0)	(-109.38)	(~ 0)	(0)	(109.38)
IA	(109.38)	(~ 0)	(109.38)	(~ 0)	(109.38)	(0)
Whole Cycle	84.286	-84.29	0	84.29	149.25	64.964

Note that the last row of the table is a total for the entire cycle. We see that we have useful work of 84.3 J and that $Q_h = 258.63$ J and $Q_c = 174.34$. We can identify Q_h and Q_c on our graph. The useful work is the difference between the work done in the two adiabatic processes.



We can see that it takes some work to make process AB happen. And recall that that is why we need a starter motor or a pull cord to get our engine started. But once the engine gets going the momentum from the piston and crank shaft keep the engine running. But still process AB requires energy input by work, so it is like a resistance for the cycle.

What is the efficiency of this cycle? We don't have a really good way with our simple view of thermodynamics to accurately account for the change in chemical potential energy involved with state I . The energy transfer in processes AI and IA are, in a sense, mechanical energy processes. So for this class it might be more fair to say that processes AI and IA have no net effect (except for refueling) and exclude their effect from Q_c and Q_h then ignoring the I processes

$$\begin{aligned}
 \eta &= \frac{W_{eng}}{Q_h} \\
 &= \frac{84.29}{149.25} \\
 &= 0.56476
 \end{aligned}$$

For a gasoline engine, it is often quoted that the efficiency is

$$\eta_{otto} = 1 - \frac{1}{\left(\frac{V_A}{V_B}\right)^{\gamma-1}}$$

Let's test this.

$$\begin{aligned}\eta_{otto} &= 1 - \frac{1}{\left(\frac{0.0005}{6.25 \times 10^{-5}}\right)^{1.4-1}} \\ &= 0.56472\end{aligned}$$

which is nearly the same. But you might guess that this is very optimistic. Efficiencies for actual engines come in at values around 20%.

